From Cross-Sections to Dynamics

Session 4

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September 10, 2026

Two ways to get change out of survey data: build a panel from cohorts, or use one that already exists.

Reading

Deaton is in your `reading/` folder on the hub, or PDF.

- Deaton (1985), "Panel data from time series of cross-sections"
- Deaton, ch. 2 §2.7 and ch. 6
- Abadie, Athey, Imbens & Wooldridge (2023), "When should you adjust standard errors for clustering?" *QJE* 138:1–35

The problem

Ghana's GLSS gives seven rounds, but almost never the same household twice. We want

$$\Delta \log c_{it} = \alpha_i + \delta_t + \beta' \Delta z_{it} + \epsilon_{it},$$

and α_i is not identified because i appears once.

Deaton's answer: give up on households and follow *cohorts*. A cohort — say, everyone born 1965–69 — is present in every round, and its membership is fixed by construction.

Pseudo-panels

Define cohort c by birth year of the head. For each (c, t) compute the weighted mean $\bar{y}_{ct}$. Then

$$\bar{y}_{ct} = \alpha_c + \delta_t + \beta' \bar{z}_{ct} + \bar{\epsilon}_{ct}$$

is a genuine panel, in $C \times T$ cells, and α_c is estimable.

The catch: $\bar{y}_{ct}$ is a *sample* mean, so it carries measurement error of order $1/\sqrt{n_{ct}}$. With $\bar{z}$ on the right-hand side that is classical errors-in-variables, and $\hat{\beta}$ attenuates.

Identification: age, cohort, and time

You cannot have all three. Since

$$\text{age} = \text{year} - \text{birth year},$$

age, period, and cohort effects are exactly collinear.
Something must be restricted:

- drop period effects (a "pure lifecycle" reading), or
- restrict period effects to sum to zero and be orthogonal to trend (Deaton & Paxson’s normalization), or
- bring in outside information on one of the three

This is not a technical nuisance to be worked around. It is a statement that the data cannot answer the question without an assumption, and the assumption should be argued for.

A genuine panel: Uganda

The Uganda National Panel Survey follows households from 2005–06 to 2018–19. With the same i in multiple t:

$$y_{it} = \alpha_i + \delta_t + \beta' z_{it} + \epsilon_{it}$$

is estimable by within transformation.

- α_i absorbs everything time-invariant — soil quality, ability, the district's road
- what remains identified is the effect of variation *within* the household over time
- and β is now a within estimate, which is a different parameter from the cross-sectional one, not a better-identified version of it

Inference

Cluster at the level of treatment assignment, not at the level of sampling.

- sampling clusters (EAs): cluster the SEs, per session 1
- if the variation you exploit is at district level, cluster on district
- few clusters ($G < 40$) $\Rightarrow$ wild cluster bootstrap
- two-way (household and time) when T is not small

Abadie et al. (2023): clustering is about *design* or *sampling*, and if neither applies, clustering is not conservative — it is wrong.

$\rightarrow$ notebook

Exercises

1. Rebuild the Ghana pseudo-panel with ten-year cohorts. How much do the lifecycle profiles change? Which conclusion is robust?
2. Implement the Verbeek–Nijman errors-in-variables correction using the within-cell variances, and report the corrected β.
3. In Uganda, quantify attrition: what fraction of the 2009–10 households appear in 2011–12? Are the leavers different in 2009–10 consumption from the stayers? What does that do to your fixed-effects estimate?